

Partial cross-location generalization of a CA1 wall-related population contrast

An exploratory longitudinal reanalysis

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Abstract

Environmental boundaries strongly influence hippocampal place fields, but it is unresolved whether wall-related remapping generalizes across physical locations. We reanalysed longitudinal calcium-imaging data from 5,413 dorsal CA1 neurons in seven mice across repeated exposure cycles containing a familiar square and nine internal-wall geometries. A cell-resolved wall-minus-open contrast estimated at one location predicted later held-out responses at the same location (mean correlation advantage 0.184; seven of seven animals). A single-tile counterfactual gave the same result (0.225; seven of seven), including after adjustment for speed, heading, time, occupancy, and spatial sampling. Crucially, a contrast estimated at a non-overlapping source seam predicted a later wall response at a target seam 25 cm away ($\bar{\rho} = 0.148$; seven of seven animals) and remained positive after behavioral adjustment ($\bar{\rho} = 0.119$). Cross-location prediction was weaker than a query-matched exact-location benchmark ($\bar{\rho} = 0.230$; shifted-minus-exact = -0.082), demonstrating partial rather than complete generalization. The correctly oriented source also exceeded an empirical equal-midpoint wrong-orientation control by 0.101 in all seven animals. A strict same-session reflected-open control favored the wall target by 0.114 in all five eligible animals, although raw-rate and fully distance-stratified comparisons were less consistent. Thus a CA1 wall-related population contrast generalizes beyond its original coordinate while retaining strong spatial dependence. The exploratory, fixed-order design does not identify whether this phenomenon reflects a wall-specific mechanism or more general spatial deformation.

Keywords: hippocampus; CA1; place cells; boundaries; population code; transfer; remapping; cognitive map

Significance statement

We tested whether a cell-by-cell CA1 response contrast associated with a wall at one location predicts a later wall response somewhere else. It did so in all seven animals, including after behavioral adjustment, but less strongly than at the original location. Spatial controls support relative wall specificity while leaving some contribution from smooth place-field structure unresolved. The central result is therefore a graded form of generalization: hippocampal remapping is neither independent across locations nor fully location invariant.

1 Introduction

Place cells relate an animal’s position to a distributed hippocampal activity pattern, yet the map is not a passive image of Euclidean space. Place fields are anchored by environmental geometry, shift when boundaries move, and can appear at characteristic distances and directions from walls (O’Keefe and Burgess, 1996; Hartley et al., 2000; Barry et al., 2006). Boundary-responsive inputs provide one mechanistic route by which local geometry could organize hippocampal representations (Lever et al., 2009). More generally, current accounts treat hippocampal-entorhinal codes as structured representations whose parts may be recombined or generalized across experiences (Stachenfeld et al., 2017; Wernle et al., 2018; Bakermans et al., 2025).

These observations leave an important distinction unresolved. A wall-related response may recur at the same coordinate despite changes elsewhere in the environment. That is *exact-location reuse*: a local feature of an otherwise changing map remains stable. A stronger possibility is *cross-location transfer*: the cell-resolved response contrast evoked by a wall at one seam predicts the response to a wall at a different seam. This second claim is not implied by ordinary place-field stability or by the existence of boundary-vector-like firing. It requires an out-of-sample test of whether the identities and signed response changes of individual cells carry across spatial coordinates.

The magnitude of cross-location prediction distinguishes three empirical possibilities. Transfer comparable to exact-location reuse would indicate weak dependence on coordinate; attenuated transfer would indicate partial generalization within a place-dependent code; and absent transfer would be consistent with responses reconstructed independently at each location. These possibilities can agree on mean rate maps while differing in the cross-registered population structure.

The dataset of Lee et al. (2025) is unusually useful for separating these possibilities. Seven mice were imaged longitudinally in dorsal CA1 while foraging in a familiar square and in nine environments made by placing internal walls on a common spatial lattice. Repeated exposure cycles provided registered cells, recurring local seams, and later held-out sessions. The design was not created for a transfer test, so the present work is exploratory and constrained by fixed environment order. Its geometry nevertheless permits three progressively stronger questions: does an estimated wall-open contrast recur at the same seam across global contexts; does an especially clean single-tile contrast recur; and does a contrast estimated at one seam predict a wall response at a neighboring, non-overlapping seam?

We find affirmative answers to all three, with an important gradient. Exact-location prediction is strongest and cross-location prediction is smaller but positive in every animal. The contribution is a specific population-level observation: a wall-related CA1 response contrast retains predictive information after a change of spatial coordinate.

2 Results

2.1 A held-out test of cell-resolved wall contrasts

We represented each candidate wall location as an oriented seam between two adjacent lattice tiles. For a registered cell, an environment, and a seam, we estimated activity in a narrow local strip and subtracted the corresponding estimate from a bracketing familiar-square session. This square residual reduced stable place preference while retaining the environment-specific local change.

Training geometries supplied two cell-wise profiles at each seam: a mean residual when the seam was blocked by a wall and a mean residual when it was open. Their difference was the estimated wall-related contrast. The neural outcome from the later target geometry was never used to form that source contrast. Prediction was scored by a rank correlation across held-out registered cells.

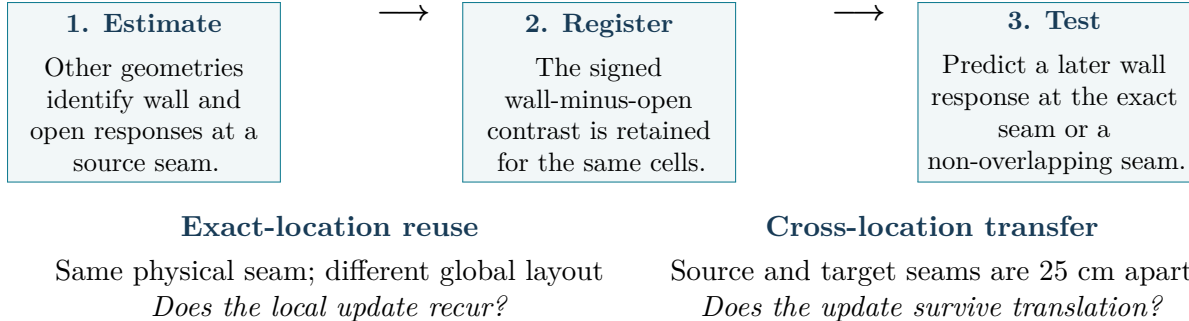


Figure 1: Analysis logic. All tests use registered-cell response vectors estimated from training geometries and evaluated on later held-out neural outcomes. Exact reuse and shifted transfer are separate estimators: the cross-location result is not a transported version of the single-tile counterfactual.

We report animal-level means and sign counts rather than treating cells or queries as independent biological replicates.

2.2 A local wall-open contrast recurs across global geometries

We first tested exact-location reuse while matching the broader wall context between training and held-out comparisons. Across 252 held-out predictions, the wall-derived profile correlated more strongly with later wall responses than did the open-derived profile. The mean wall-minus-open correlation advantage was 0.184 and was positive in all seven animals (Fig. 2A). The sign was also positive in all animals when horizontal and vertical seams were evaluated separately.

This analysis establishes more than a population-average increase near walls. The predictive object is a vector indexed by registered cell identity. Cells that contributed positively or negatively to the estimated local contrast tended to do so again when the same seam was encountered in another global geometry. However, the seam itself was unchanged, so this result alone does not show that the update can move through space.

2.3 A one-tile counterfactual sharpens exact-location reuse

Two environments in the sequence, denoted u and o , differed only in whether one lattice tile was blocked. Their contrast therefore isolates one local wall state more cleanly than comparisons among complex layouts. We estimated the $u - o$ cell-wise contrast at that focal seam and tested it in held-out i , l , and t geometries that independently instantiated wall or open states there.

Across 46 held-out predictions, the estimated vector correlated with wall outcomes ($\bar{\rho} = 0.197$) but not open outcomes ($\bar{\rho} = -0.028$), giving an advantage of 0.225, positive in all seven animals (Fig. 2A). The raw-rate version gave a similar advantage (0.243). A model that adjusted local bin estimates for speed, heading, elapsed time, and occupancy retained the result, both with time included (0.225) and omitted (0.229).

The effect was more evident in the nearest independently eligible distance band, 2.5–7.5 cm from the seam (mean advantage 0.233, seven of seven animals), than in a 17.5–22.5 cm band (0.069, six of seven; Fig. 2D). These bands had separately defined supports and are therefore a descriptive localization comparison, not a paired inferential contrast.

The fixed sequence imposes an important caveat: u always followed o . A generic same-seam, unchanged-state later-minus-earlier placebo was opposite in sign (mean -0.042 , zero of seven positive), whereas the actual $u - o$ contrast was positive (mean 0.195, seven of seven) and remained

so after the nuisance adjustment (mean 0.212). This argues against a generic drift explanation, but it cannot remove an environment-specific order effect because wall state and order were never randomized.

2.4 A wall-related population contrast generalizes across locations

We next asked the stronger question. For each eligible target, we estimated a wall-minus-open vector at a distinct source seam with the same signed wall normal. The source and target seams were 25 cm apart, their local analysis bins did not overlap, and source training sessions were disjoint from the target session. The nearest source-target bins were 5 cm apart for tangential translations and 15 cm apart for normal-axis translations. The test used a separate multi-geometry estimator rather than the $u - o$ contrast.

The translated source vector predicted the later target-wall residual with mean correlation 0.148, positive in all seven animals (Fig. 2B). It also outperformed a source-open predictor by 0.152, and was more aligned with the target-wall response than with a training-derived target-open profile by 0.216, again positive in every animal. Thus the result cannot be summarized merely as two wall locations having high activity: a signed, cell-resolved contrast estimated at one coordinate carried information about another.

Transfer was incomplete. On exactly the same eligible queries, cells, and support, an exact-location effect predicted the target response with mean correlation 0.230. The shifted-minus-exact difference was -0.082 , and only one of seven animals favored the shifted predictor. The data therefore show partial generalization with substantial dependence on the original coordinate.

2.5 Behavioral and spatial controls define the phenomenon

Because animals may run differently near internal barriers, we re-estimated local neural response maps with covariates for speed, first and second circular harmonics of heading, elapsed time, and spatial sampling. The behavior-adjusted translated effect remained positive in all seven animals ($\bar{p} = 0.119$). Its specificity against the training-derived target-open profile was 0.172, and its advantage over a source-open predictor was 0.138, both seven of seven. Omitting elapsed time changed these values by less than 0.003.

We next compared the correctly oriented source with empirical source seams at the same 25 cm target-midpoint distance but with an incorrect signed wall normal. This control retains the observed neural maps, registered-cell identities, target response, and source-target lag while breaking the correct wall relation. The correct-minus-incorrect advantage was 0.101, positive in all seven animals. Complete enumeration of the 2^7 animal-level sign assignments placed the observed mean in the upper 1/128 of that descriptive null. The result remained positive in every leave-one-animal-out summary.

This pooled comparison does not eliminate spatial continuity. When restricted to tangential translations with both a 25 cm midpoint distance and a 5 cm nearest-strip lag, the correct-orientation advantage was only 0.040, positive in three of six animals (sign-flip tail fraction 0.25). Both correctly and incorrectly oriented tangential sources predicted the target. Thus the pooled null shows that the full effect is not an arbitrary equal-midpoint match, but the most tightly distance-matched stratum does not identify a general orientation-specific rule.

The registered-wall and target-wall strips could nevertheless share generic nearby place-field structure. We therefore defined a stricter within-session control before inspecting its neural endpoint in the project workflow. For each wall target, a mirror-open seam in the same target session was matched for translation distance, axis, signed normal, relative bins, and registered cells. The

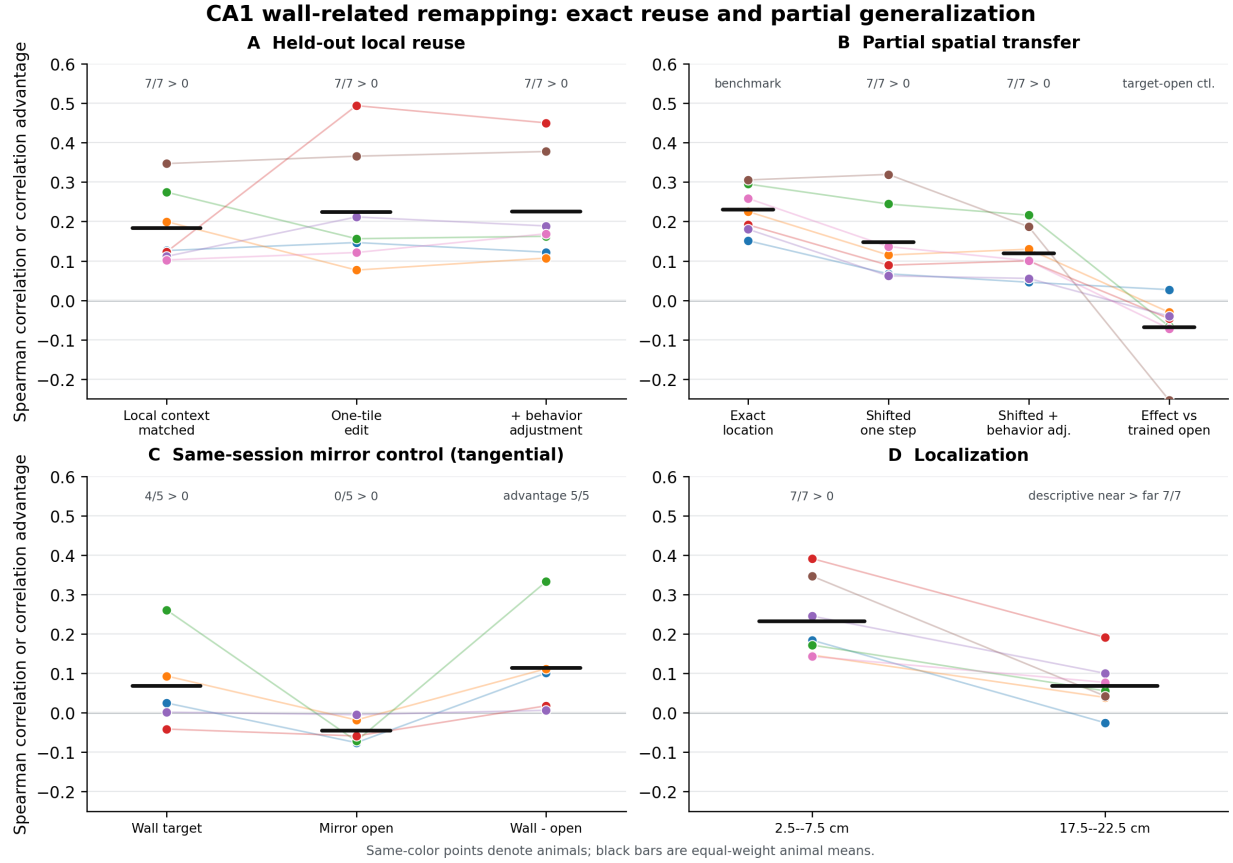


Figure 2: Exact reuse, shifted transfer, and key controls. **A**, wall-minus-open prediction advantages at the same seam, including the single-tile contrast and its behavior-adjusted estimate. **B**, the query-matched exact-location benchmark, the translated source-effect prediction, its behavior-adjusted estimate, and its specificity against a training-derived target-open profile. **C**, the same-session mirror-open control on its supported paired subset: correlations for the wall and mirror-open targets and their wall-minus-open advantage. Only five animals had eligible paired support. **D**, descriptive near and far bands for the single-tile test; the bands were separately eligible and should not be read as a paired estimate. Points are animal means; bars show the across-animal mean.

geometry permits this match only for tangential translations. Although 104 labeled configurations were feasible, paired-support gates retained 72 triplets in five animals; two animals had no admissible control.

On this restricted subset, the globally demeaned source vector favored the wall target over the mirror-open target by 0.114, positive in all five eligible animals (Fig. 2C). Absolute source-to-wall correlation was positive in four of five animals ($\bar{p} = 0.068$), and a direct within-session wall-minus-open difference-vector score was positive in four of five ($\bar{p} = 0.077$). The raw-rate primary was positive in only three of five. The exact animal-level sign-flip tail fraction for the demeaned advantage was 1/32, and every leave-one-animal-out mean remained positive. The strongest justified reading is narrow: the transferred contrast distinguished a wall target from an equally displaced open location in the supported demeaned tangential cases. It does not establish the full seven-animal effect, normal-axis transfer, or raw-rate specificity.

A coherent source-cell identity permutation further showed descriptive upper-tail fractions below 0.01 in five animals and values of 0.069 and 0.060 in the other two. Registered-cell correspondence

Table 1: Evidence for partial cross-location generalization. Values are means across animal-level estimates. Sign counts are descriptive consistency summaries.

Test	Mean	Sign	Supported interpretation
Context-matched exact seam	0.184	7/7	Local wall-open population contrast recurs across layouts.
Single-tile exact seam	0.225	7/7	Exact-location reuse survives a cleaner local counterfactual.
Behavior-adjusted single tile	0.225	7/7	Measured speed, heading, time, and sampling do not explain reuse.
Shifted source effect	0.148	7/7	A cell-resolved wall contrast predicts a different location.
Query-matched exact effect	0.230	7/7	Exact reuse exceeds shifted transfer.
Behavior-adjusted shifted effect	0.119	7/7	Measured behavior does not eliminate transfer.
Correct minus wrong orientation at 25 cm	0.101	7/7	The pooled effect exceeds an equal-midpoint empirical spatial null.
Mirror-open advantage	0.114	5/5	Demeaned tangential subset favors a wall over an equally displaced open seam.

therefore contributes to the prediction, although the effect is not uniformly strong. Additional exploratory analyses of direction, composition, experience, and a former-barrier endpoint are reported in Supplementary Section 5 rather than treated as tests of the central phenomenon.

3 Discussion

The central result is partial cross-location generalization in a hippocampal spatial code. A cell-resolved wall-minus-open response estimated from one group of geometries predicted a later response at the same seam, and a contrast estimated at a different seam predicted a later target-wall response. The cross-location prediction was positive in every animal, survived adjustment for measured behavior, and was weaker than the matched exact-location prediction. Local boundary-related remapping is therefore neither independent across coordinates nor fully location invariant.

The finding goes beyond the established observation that boundaries influence place fields. Boundary-vector models explain how fields can arise from inputs tuned to distance and direction from boundaries (Hartley et al., 2000; Barry et al., 2006), and barrier manipulations have shown local field formation, duplication, and geometry-dependent remapping (Rivard et al., 2004; Lee et al., 2025). Those results make some recurrence plausible. They do not by themselves demonstrate that a signed vector of responses, indexed by the identities of longitudinally registered CA1 cells, generalizes from one physical seam to another under held-out evaluation.

The attenuation relative to exact reuse is equally informative. It shows that the response is not reconstructed independently at every seam, yet that most predictive structure remains tied to the original location. This intermediate organization is the empirical result; no latent operator or particular biological decomposition is required to state it.

Several alternative explanations remain. First, smooth or repeated place fields can correlate at nearby seams. The non-overlapping strips, square residuals, target-open specificity, equal-midpoint wrong-orientation null, and mirror-open control constrain this account. They do not eliminate it: the fully distance-matched tangential orientation contrast was inconsistent, and the mirror control was limited to five animals and demeaned tangential translations. Second, unmeasured behavior such as attention, wall inspection, acceleration, or route choice could contribute despite adjustment for measured kinematics. Third, the fixed environment sequence leaves environment-specific order confounded with the single-tile contrast. Fourth, registration and activity thresholds select cells

stable enough to be observed across sessions. Fifth, the identity permutation is not an ideal exchangeability test for correlated place-cell populations.

Exploratory analyses of composition, experience, and a former-barrier endpoint were either underpowered or unstable and are reported separately in the Appendix. They do not alter the positive cross-location result and should not be interpreted as broad tests of hippocampal composition, learning, or innateness.

The decisive next experiment is compact. Use a symmetric lattice arena with four equivalent candidate wall seams. Randomly assign source and target locations, wall orientation, insertion or removal, and session order within animal. Estimate a source wall-open contrast, then test exact reuse, translated reuse, and a distance-matched open target in the same later session. This design would distinguish wall-specific generalization from smooth spatial deformation while removing the fixed-order confound. The present effect sizes provide realistic targets: an exact-location correlation near 0.23, shifted correlation near 0.12–0.15, and a shifted-minus-exact penalty near 0.08.

4 Materials and methods

4.1 Dataset and analysis cohort

We reanalysed the public longitudinal dorsal-CA1 calcium-imaging dataset of [Lee et al. \(2025, 2024\)](#). The tracked analysis cohort contains seven mice, 5,413 neurons, 207 sessions, a familiar outer square, and nine internal-wall geometries arranged on a common lattice. Cell identities were registered across sessions in the source dataset. The original study describes surgery, imaging, behavior, preprocessing, registration, and animal procedures. No new animal experiments were performed.

4.2 Spatial representation and oriented seams

The arena was expressed in a common allocentric coordinate system. A candidate internal-wall location was represented as an oriented seam $s = (p \rightarrow q)$ between adjacent lattice tiles, with a signed normal from tile p to tile q . The binary state $z_{e,s}$ indicated whether environment e blocked that adjacency. Local strips were partitioned into 5 cm spatial bins at defined distances from the seam. Analyses used only bins and registered cells satisfying the relevant occupancy, activity, and common-support gates. Source and target strips in the cross-location analysis shared no 5 cm bin.

4.3 Local response vectors and square residuals

Let $r_{i,e,s}$ denote the local response estimate for registered cell i in environment e at seam s , averaged over eligible spatial bins after the analysis-specific preprocessing. The corresponding bracketing-square estimate was $r_{i,\square(e),s}$. The primary environment-specific response was

$$\delta_{i,e,s} = r_{i,e,s} - r_{i,\square(e),s}.$$

For globally demeaned variants, each vector’s common-cell mean was removed before correlation. This suppresses population-wide rate shifts and makes the score depend on relative cell identities.

4.4 Training profiles and held-out scoring

For a target query (e^*, s^*) , training environments were selected without using the neural response of e^* . At a source seam s , the wall and open profiles were

$$\boldsymbol{\mu}_{W,s} = \text{mean}_{e \in \mathcal{T}: z_{e,s}=1} \boldsymbol{\delta}_{e,s}, \quad \boldsymbol{\mu}_{O,s} = \text{mean}_{e \in \mathcal{T}: z_{e,s}=0} \boldsymbol{\delta}_{e,s},$$

computed on the common registered-cell support. The estimated source effect was $\mathbf{d}_s = \mu_{W,s} - \mu_{O,s}$. The held-out outcome was δ_{e^*,s^*} . Scores were Spearman correlations across registered cells. Wall-open advantages compare correlations within the same query and cell support.

The exact-location estimator set $s = s^*$. Its context-matched version restricted training comparisons so the broader internal-wall context was balanced between the wall and open profiles. The reported exact-location benchmark for shifted transfer was recomputed on the exact queries, sessions, cells, and local supports eligible for the shifted analysis.

4.5 Single-tile counterfactual

The u and o geometries differed at one lattice tile: u blocked tiles $\{4, 5\}$, whereas o blocked only tile $\{4\}$. We formed the cell-wise $u - o$ contrast at the focal seam and evaluated wall and open outcomes at that seam in held-out i , l , and t geometries. Near (2.5–7.5 cm) and far (17.5–22.5 cm) bands were screened separately, so their difference is descriptive. For the order-placebo analysis, we formed later-minus-earlier contrasts at unchanged seams with the same processing and compared their signs with the actual $u - o$ contrast.

4.6 Cross-location transfer

For each eligible target seam, the source seam was a non-overlapping lattice seam 25 cm away with the same signed wall normal and admissible wall/open training support. Source training and target outcome sessions were disjoint. The primary shifted score was

$$\rho_{\text{shift}} = \rho_S(\mathbf{d}_{s_{\text{source}}}, \delta_{e^*, s_{\text{target}}}).$$

Secondary specificity scores compared the source effect with a training-derived open profile at the target and compared source-wall with source-open predictors. The exact-location benchmark used $\mathbf{d}_{s_{\text{target}}}$ on the identical held-out query support. Stored-map and behavior-adjusted audits agreed on 405 target queries and 658 source pairs, including their session and seam identities and their counts of common cells and bins. That audit did not hash the complete cell-ID or bin-coordinate arrays.

4.7 Behavior-adjusted response estimates

We estimated activity jointly with spatial-bin indicators and nuisance covariates for instantaneous speed, sine and cosine of heading, second heading harmonics, and elapsed time. Spatial effects were evaluated at a common reference covariate distribution calibrated from the first familiar-square session. The no-time sensitivity omitted elapsed time but retained the other covariates. Target-session activity was used to estimate nuisance coefficients for that target map, but target neural outcomes never entered source-profile training.

4.8 Same-session mirror-open control

Before inspecting the neural endpoint within the project workflow, we defined a mirror-open seam in the same target session as each target wall. It matched the source-wall pair in translation distance, translation axis, signed normal, relative spatial bins, and registered cells. Both source-to-target nearest-bin distances were 5 cm. Only tangential translations admitted this exact geometric pairing. Primary scoring used globally demeaned source and target vectors. Eligibility required a complete source, wall-target, and open-target triplet on common support.

4.9 Spatial-null calibration

We calibrated two empirical spatial controls at the animal level. The first compared correctly oriented source seams with incorrectly oriented sources at the same 25 cm target-midpoint distance. A tangential stratum additionally matched translation axis and the 5 cm nearest-strip lag. The second compared the wall target with the same-session reflected-open target described above. Both controls retain the observed neural maps and therefore their place-field smoothness. For each paired animal-level contrast, we enumerated all 2^N sign assignments and calculated the fraction of null means at least as large as the observed mean. These are descriptive exact sign-flip calibrations, not confirmatory population p -values.

4.10 Cell-identity permutation

For each animal, we generated 999 coherent random mappings of source-cell identities while preserving query geometry, target vectors, support sizes, and the mapping across all queries in that permutation. The descriptive upper-tail fraction was $(1 + n_{\text{null} \geq \text{observed}})/(1000)$. Correlations and registration constraints limit exchangeability; the fractions were used as identity-specificity diagnostics rather than inferential population p -values.

4.11 Direction, composition, and experience analyses

Direction controls compared eligible source-target pairs with same or opposite signed wall normals while documenting translation axis, distance, overlap, and layout matching. No single comparison simultaneously matched all of these features across the full cohort.

For composition, we identified complete wall-state factorials in which two single-wall effects and their joint state could be estimated. The additive prediction was the sum of the two single effects. It was compared with an oracle best-single-component predictor using correlation and normalized prediction error. Only two repetitive factorials and five eligible animals were available.

Experience analyses contrasted transfer changes across exposure blocks using difference-in-differences variants. Because geometry, order, and exposure were not independently randomized, these estimates were treated as exploratory sensitivity analyses and not as evidence for learning.

4.12 Independent former-barrier analysis

We used the public neutral-barrier data from [Blair et al. \(2023b\)](#) to test whether the former barrier center retained an excess neural similarity relative to matched control locations after removal. The primary endpoint was a Pearson correlation among active cells, averaged as a former-barrier center-minus-control contrast. Its definition was written before the neural endpoint was run in the local project workflow. Spearman scoring and leave-one-animal-out summaries were sensitivities.

4.13 Statistical reporting and reproducibility

Animals were treated as the biological units. We report animal-level means, sign counts, query counts, and explicitly labeled descriptive permutation fractions. No cell-level or query-level observation was presented as an independent animal replicate. The analyses are exploratory: neither the original experiment nor the present reanalysis was preregistered for the cross-location hypothesis.

All figure values derive from tracked machine-readable source files. The repository contains scripts, eligibility ledgers, result summaries, design locks, and an inspectable Git history. The principal manuscript figure is generated from its tracked source-data CSV.

5 Supplementary exploratory analyses

5.1 Direction and two-wall composition

Direction controls did not identify a general orientation rule. Same- versus opposite-signed normals for tangential translation favored the same direction in three of six eligible animals. A normal-axis comparison against opposite-facing or overlapping alternatives was positive in four of seven. A less controlled same- versus opposite-away comparison was positive in all seven, but it also changed strip distance and layout. These analyses support transfer between selected corresponding seams, not rotation- or direction-invariant transport.

Only two complete two-wall factorials were available, both involving the same bit and donut layouts. In five eligible animals, the demeaned additive predictor reached mean correlation 0.153, below an oracle best-single-component predictor at 0.178. The additive advantage was -0.024 and positive in one of five animals. This sparse support provides no positive evidence for linear two-wall composition but cannot establish general non-compositionality.

5.2 Experience and former-barrier endpoints

A transfer-by-experience difference-in-differences estimate was positive under one stored-map specification (mean 0.045, four of four eligible animals), but raw-rate and stricter sensitivity analyses were unstable. Environment order, exposure number, and geometry were not independently varied, so the data do not identify experience-dependent stabilization or innateness.

In an independent neutral-barrier dataset (Blair et al., 2023b), the locked Pearson-active former-barrier center-minus-control endpoint was negative in all leave-one-animal-out summaries (animal mean -0.183 ; one of six animals positive). A Spearman sensitivity was near zero (mean 0.001; three of six positive). We therefore found no evidence for a persistent local representational scar under that endpoint.

Data and code availability

The CA1 source data are available through the archive cited by Lee et al. (2024). The independent miniscope data are available through the archive cited by Blair et al. (2023a). Analysis code, derived source tables, audit reports, and manuscript sources are publicly available at <https://github.com/asim-v/ca1-wall-update-transfer>. The public release excludes the original raw recordings, which remain available from their source archives.

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Author contributions

J.E.B.S. conceived the study, developed and performed the analyses, interpreted the results, prepared the visualizations, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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